Quadratic Coefficients

Two Variables Example

$$(x_1\ x_2)\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = (a_{11}x_1 + a_{21}x_2 \quad a_{12}x_1 + a_{22}x_2)\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$= (a_{11}x_1 + a_{21}x_2)x_1 + (a_{12}x_1 + a_{22}x_2)x_2 = a_{11}x_1^2 + a_{21}x_1x_2 + a_{12}x_1x_2 + a_{22}x_2^2$$

$$= a_{11}x_1^2 + (a_{21} + a_{12})x_1x_2 + a_{22}x_2^2 \Rightarrow \text{Quadratic form}$$

$a_{11} \Rightarrow$ Coefficient of x_1^2

$a_{21} + a_{12} \Rightarrow$ Coefficient of x_1x_2

$a_{22} \Rightarrow$ Coefficient of x_2^2

$a_{12} = a_{21} = \frac{1}{2}$ Coefficient of x_1x_2

$$\begin{matrix} & \overset{x_1}{} & \overset{x_2}{} \\ x_1 & x_1 & x_2 \\ x_2 & & \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \Rightarrow$$

coef of x_1^2	0.5 coef of x_1x_2
0.5 coef of x_1x_2	coef of x_2^2